

Habitual intake of dietary flavonoids and risk of Parkinson disease



Objective: To prospectively examine whether higher intakes of total flavonoids and their subclasses (flavanones, anthocyanins, flavan-3-ols, flavonols, flavones, and polymers) were associated with a lower risk of developing Parkinson disease (PD). Methods: In the current analysis, we included 49,281 men in the Health Professional Follow-up Study and 80,336 women from the Nurses' Health Study. Five major sources of flavonoid-rich foods (tea, berry fruits, apples, red wine, and orange/orange juice) were also examined. Flavonoid intake was assessed using an updated food composition database and a validated food frequency questionnaire. Results: We identified 805 participants (438 men and 367 women) who developed PD during 20–22 years of follow-up. In men, after adjusting for multiple confounders, participants in the highest quintile of total flavonoids had a 40% lower PD risk than those in the lowest quintile (hazard ratio [HR] = 0.60; 95% confidence interval 0.43, 0.83; p trend = 0.001). No significant relationship was observed in women (p trend = 0.62) or in pooled analyses (p trend = 0.23). In the pooled analyses for the subclasses, intakes of anthocyanins and a rich dietary source, berries, were significantly associated with a lower PD risk (HR comparing 2 extreme intake quintiles were 0.76 for anthocyanins and 0.77 for berries, respectively; p trend < 0.02 for both). Conclusions: Our findings suggest that intake of some flavonoids may reduce PD risk, particularly in men, but a protective effect of other constituents of plant foods cannot be excluded.